Physics - Graduate Level Assessment 22 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A chain of mass m per unit length and total length l lies in a pile on the floor. At time t = 0 a force P is applied and the chain is raised with constant velocity v. Find P(t).
   (a) P = mv(t + g)
   (b) P = m(v + $gt)^2$
   (c) P =mgvt+ mv^2 =mv(v +gt)
   (d) P = mv^2 - mgvt
   (e) P = mv + gt

2. How much heat is produced in 5 minutes by an electric iron which draws 5 amperes from a 120-volt line?
   (a) 72,000 calories
   (b) 80,000 calories
   (c) 36,000 calories
   (d) 65,000 calories
   (e) 48,000 calories

3. The two moons of Mars are called ...
   (a) Deimos and Triton
   (b) Phobos and Demos
   (c) Triton and Deimos
   (d) Phobos and Deimos
   (e) Tritos and Desmos

4. The electric flux through a spherical surface is $4.0 \times 10^4 N \cdot m^2/C$. What is the net charge enclosed by the surface?
   (a) 2.65 e-07
   (b) 7.10 e-07
   (c) 2.30 e-07
   (d) 8.45 e-07
   (e) 5.67 e-07

5. What is the illumination 5 feet from a 75 candlepower lamp?
   (a) 5 foot-candles

(b) 1 foot-candle

(c) 50 foot-candles

(d) 15 foot-candles

(e) 20 foot-candles

True / False (10 marks)

Answer True or False

6. True or False: The force applied by the pitcher to the baseball is 10 pounds, which corresponds to a higher mass or lower acceleration than stated.

7. True or False: The magnifications for a thin lens with a focal length of 50 mm at object distances of 20 mm, -100 mm, and +100 mm are (4/3), (1/5), and 0.

8. True or False: The work done against friction when dragging a 150 kg mass over 10 meters with a coefficient of friction of 0.30 is 5000 joules.

9. True or False: The focal length of the lens in a slide projector designed for 35-mm slides projecting at a distance of 10 ft is 2.69 inches.

10. True or False: The approximate atomic weight of natural boron, consisting of 20% $^{1}0Band80\%^{1}1B,is20onthescaleof16unitsfor^{1}6O$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the given x-acceleration function a_x = 2.0 m/s^2 - (0.10 m/s^3) t for Sally's 1965 Mustang. Determine the distance X at which the car reaches its maximum x-velocity, and explain your reasoning.

12. Calculate the total heat required to transform 25 kg of ice from -10 $^{t}osteamat100^{,}detailingeachphasec$

End of Paper